



FEATURES BASED ON LOCATION OF BLEED

00:13:30

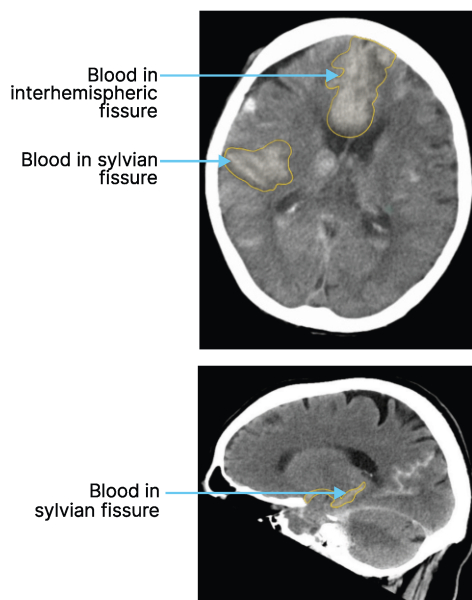
FEATURES

- | | | | |
|--|--|---|---|
| <ul style="list-style-type: none"> • Anterior communicating artery <ul style="list-style-type: none"> ◦ Abulia (lack of initiation/ drive) ◦ Aphasia (involvement of Broca's area) | <ul style="list-style-type: none"> • Posterior communicating artery <ul style="list-style-type: none"> ◦ Ptosis (compression of CN III) ◦ CN III palsy | <ul style="list-style-type: none"> • Internal carotid artery <ul style="list-style-type: none"> ◦ Present in cavernous sinus ◦ CN VI palsy | <ul style="list-style-type: none"> • Occipital headache - PICA, AICA • MCA - Temporal headache |
|--|--|---|---|

WORK UP

00:16:39

- **Initial investigation - NCCT Head**
 - Blood in Sylvian fissure/interhemispheric fissure
- **Next best investigation - Invasive Cerebral angiography > CT angiography**
 - Demonstrate → cause, locations
 - T/t endovascular coiling
 - Mirror berry aneurysm → 20%
- **Guarded LP**
 - Guarded LP → Bloody CSF
 - RBC lysis Xanthochromic CSF
 - Appears > 12-24 hours, Peak at 48 hours
 - May persists up to 4 weeks
- ECG
 - **T wave inversion**
 - Reason - extreme pain → catecholamine surge
 - vasoconstriction of coronary arteries
- Serum electrolytes - ↓ Na²⁺



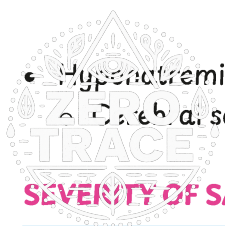
Important Information

- Bloody CSF is seen in Traumatic LP/ SAH
 - Traumatic LP (Injury to epidural venous plexus)- **3 tube test** differentiates between SAH and Traumatic LP
 - Traumatic LP → subsequent CSF samples will be clear
 - SAH → uniform bloody CSF

DELAYED NEUROLOGICAL DEFICITS

00:25:57

- Re-rupture - Strict bed rest
- Hydrocephalus
 - Mitigation - external ventricular drainage
- Delayed cerebral ischemia (leading cause of delayed mortality)
 - Delayed mortality - vasospasm of feeder vessel
 - Investigations - Transcranial doppler (noninvasive), CTA/ CTP
 - Primary prevention - **Nimodipine** (prevents spasm of feeder vessels)



- Hyponatremia
• Cerebral salt wasting syndrome (Hypovolemic hyponatremia)

SEVERITY OF SAH

GRADE	HUNT-HESS SCALE	WORLD FEDERATION OF NEUROSURGICAL SOCIETIES (WFNS) SCALE
1	Mild headache, normal mental status, no cranial nerve or motor findings	GCS score 15, no motor deficits
2	Severe headache, normal mental status, may have cranial nerve deficit	GCS score 13-14, no motor deficits
3	Somnolent, confused, may have cranial nerve or mild motor deficit	GCS score 13-14 with motor deficits
4	Stupor, moderate to severe motor deficit, may have intermittent reflex posturing	GCS score 7-12, with or without motor deficits
5	Coma, reflex posturing or flaccid	GCS score 3-6, with or without motor deficits

T/t

00:30:38

- Endovascular coiling
 - Gives structural support to Blood vessels
 - Minimizes rebleeding
 - TOC - Endovascular coiling > Aneurysmal clip
- Aneurysmal clip - Craniotomy to be done
- Secure airway- 1st step in unconscious brain injury
- BP control - BP > 220/130 mmHg + end organ damage
 - Nicardipine > labetalol
- Prophylactic anticonvulsant - Phenytoin
- Vasospasm management - Nimodipine (initial drug)
 - Intra-arterial vasodilator, balloon dilatation for persistent vasospasm
- Hyponatremia causing seizures - 3% saline
- Prevention of DVT -
 - Pneumatic compression
 - Enoxaparin after 48 hours



- Q. 30-year-old lady presents with sudden onset headache and nuchal rigidity for past 12 hours. NCCT head shows blood in Sylvian fissure. BP on admission 160/100 mmHg. What is the next step in management of this patient?
- a. Cerebral angiography
 - b. Mannitol
 - c. Lower BP with Nicardipine
 - d. Stat dose of Nicardipine

Ans: (a)

- Q. Which of the following is not correct about findings seen in SAH?
- a. ECG shows T wave inversion
 - b. Dyselectrolytemia
 - c. Seizures
 - d. M/c cause is rupture of Berry aneurysm

Ans: (d)



TRANSIENT ISCHEMIC ATTACK AND STROKE INTERVENTIONS

INTRODUCTION

00:00:53

- 4 million neurons, 12 million brain cells, 15 million synapses die every minute during a stroke
- For every 1-hour delay in T/t of acute ischemic stroke, brain age ↑ by **3.6 years**

Extra Mile

- Types of stroke
 - Acute Ischemic Stroke (AIS): ~85%
 - Hemorrhagic Stroke: ~15%
 - Venous Stroke: < 1%
- M/c Causes:
 - Acute Ischemic Stroke: **Non-valvular atrial fibrillation (NVAf)**
 - Hemorrhagic Stroke: **Hypertensive crisis**
- Door-to-Imaging Time (DIT): Should be **<25 minutes**

IDENTIFICATION

Mnemonic: **BE FAST**

- In any patient with atherosclerotic risk factors presenting with sudden onset of:
 - **B**alance issues
 - **E**ye deviation
 - In Cortical stroke: eyes deviate towards the side of lesion
 - **F**acial droop
 - **A**rm weakness
 - **S**peech impairment: Global aphasia
 - **T**ime to call 102/108

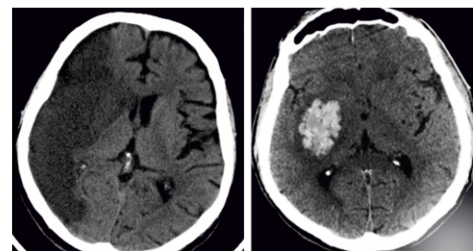


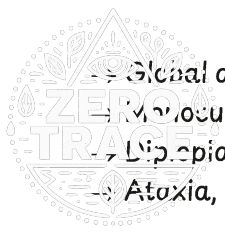
Important Information

- Preferred Antihypertensive Drugs in Hypertensive Crisis:
 - Cardiac emergency (LVF): **Nitroglycerin (NTG)**
 - Neurological emergency (Hemorrhagic stroke): **Nicardipine**

WHEN TO SUSPECT STROKE

- Any patient with sudden onset of:
 - **Altered level of consciousness:**
 - ↑ ICP → impairs the reticular activating system → altered consciousness
 - **Neurological deficits:**
 - Monoparesis / Hemiparesis
 - Hemisensory loss





- Global aphasia: typically **Left MCA** involvement
- Monocular/binocular visual loss: ophthalmic artery involvement
- Diplopia, dysarthria, facial droop
- Ataxia, vertigo

TRANSIENT ISCHEMIC ATTACK

00:11:20

A small clot temporarily blocks a brain blood vessel



Triggers activation of the body's natural anticoagulant systems
(**Protein C, Protein S, Antithrombin III**)



Clot is lysed & blood flow is restored



Neurological symptoms resolve as perfusion returns

- Usually, improvement occurs within **1 hour**
- TIA symptom resolution cut-off: **within 24 hrs**

CASE SCENARIO 1

- 60-year-old male smoker with Hypertension and T2DM presents with sudden onset symptoms followed by **spontaneous improvement**
 - Transient loss of vision (Amaurosis fugax)
 - Thrombus in **Ophthalmic artery**
 - Right arm weakness, speech impairment, facial droop
 - Thrombus in **MCA territory**
 - Sudden onset Paraparesis / Monoparesis
 - **ACA involvement**
 - Nasal twang of voice / Nasal regurgitation of fluids
 - **Vertebral artery thrombosis** → cranial nerve palsies
 - **CN IX** involvement → nasal twang, nasal regurgitation
 - **CN X** involvement → dysarthria or anarthria
 - Hemisensory symptoms: Pins-and-needles/paresthesia
- If any symptom → resolves spontaneously within 24 hours → considered TIA
- TIA → **highest stroke** risk in next **48 hrs**

ABCD² SCORE

- Purpose: **Predicts the chance of stroke development** within the **next 3 months** after a TIA
- **ABCD² score**
 - **A** - Age > 60 years: **1 point**
 - **B** - BP > 140/90 mmHg: **1 point**
 - **C** - Clinical Features
 - Arm weakness: **2 points**
 - Speech impairment: **1 point** -
Duration of symptoms
 - 10-59 minutes: **1 point**
 - > 60 minutes: **2 points**
 - **D** - Diabetes mellitus: **1 point**



- Total = 7 points
- $\text{CHA}_2\text{DS}_2\text{-VASc} \geq 2 \rightarrow \sim 22\% \text{ risk of stroke occurrence}$

Important Information

- **ABCD² score** predicts stroke risk → not diagnostic
- To confirm diagnosis of stroke: **NIHSS** (National Institutes of Health Stroke Scale) is used

WORK-UP

- **ECG**
 - Rule out **atrial fibrillation**
- Echocardiography
 - Types: Transthoracic/Transesophageal Echocardiography
 - Assess → Left atrial appendage clots (LAA)
- **Doppler/duplex of carotid artery**
 - To detect atherosclerotic narrowing of ICA
- Lipid profile → ↑ LDL & ↓ HDL
 - ↑ LDL and ↓ HDL
- **Complete Metabolic Profile**
 - Checked Random Blood Sugar values

TREATMENT

00:32:26

Mnemonic: **ABCDX**

Antiplatelet:

- **Dual therapy** (Aspirin + Ticagrelor)
 - Loading dose on day 0
 - Continue low-dose: **21-30 days**
 - **Stop** d/t **GI bleeding side effects**
- Monotherapy → **Aspirin** after 30 days
 - Low dose enteric coated **aspirin 75mg**
- **BP control** → **ACE inhibitors** & diuretics/**thiazides**
- **Cholesterol control**
 - Target LDL < **70 mg/dL**
 - High-intensity statins: Atorvastatin 40-80 mg
- **Carotid Endarterectomy Indications**
 - High-grade ICA stenosis
 - Recent hemispheric symptoms
 - Low institutional morbidity /mortality
- **DM control**
 - Start insulin if HbA1C >10%
- Anticoagulation
 - **DOAC** → Apixaban/ Rivaroxaban/ Dabigatran
 - Indication: Nonvalvular **atrial fibrillation** with **CHADS₂-VASc score >2**
(CHADS₂-VASc score: to decide the need for **OAC**)
 - Warfarin (for Valvular Atrial Fibrillation)
 - Indications: RHD with mitral stenosis and LAA clots



-> Mechanical heart valves

INR: 2.5-3.5 in Mechanical heart valves

ACUTE ISCHEMIC STROKE (AIS)

00:41:05

CASE SCENARIO 2

60-year-old male smoker with hypertension and T2DM got up at 6 a.m., brushed his teeth, and got ready for a walk. At 6:30 a.m., his wife noticed sudden-onset facial droop and right arm weakness. At 7:30 a.m., you, the resident, examine him; BP on admission is 170/100 mm Hg. What are the steps to be followed during an acute stroke consultation?

STEP 1

- Airway Assessment, BP & time last seen normal
 - Thrombolysis is C/I if BP > 185/110 mm Hg → ↓ BP using CCBs
 - Time last seen normal: helps determine the thrombolysis window period
→ Window period for thrombolysis with Alteplase in STEMI: 12 hours

THROMBOLYSIS WINDOW IN AIS (ALTEPLASE)

Ideal: <4.5 hours

Upper limit: <6 hours

STEP 2

- Calculate NIH Stroke Scale score:
 - If value is >5 → Stroke

STEP 3

- Perform neuroimaging: NCCT head + blood work including RBS
 - Door-to-Imaging time: < 25 mins

Important Information

- In acute ischemic stroke (AIS), initial CT scan may appear normal
 - Ischemic changes typically take up to 6 hours to become visible
- CT scan is done to rule out hemorrhagic stroke (findings appear immediately)
- Thrombolysis is contraindicated in hemorrhagic stroke

STEP 4A

- <6 Hours since symptom onset
 - NCCT Head shows evidence of hemorrhage
 - Perform CTA to demonstrate Large Vessel stroke/occlusion: M1, M2, ICA, BA
 - Check eligibility for thrombolysis:
 - IV Alteplase (0.9 mg/kg) or Tenecteplase
 - Alteplase: 10% bolus, remaining as an infusion over 1 hour
 - Tenecteplase: Given as a bolus

Outcomes

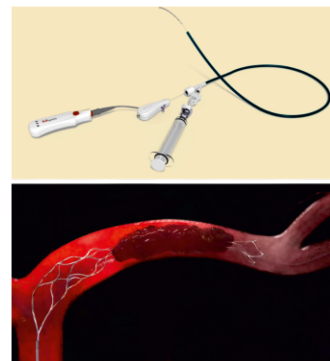
- IV alteplase success rate ≈30% → transfer to cath lab for cerebral angiography + thrombectomy if feasible
- If thrombolysis is C/I → Perform thrombectomy directly



- 6-24 Hours of Symptom Onset
 - NCCT: No evidence of hemorrhage
 - CTA/CTP: Favourable perfusion → perform thrombectomy

Indications for Thrombectomy

- A. AIS with 6-24 hrs onset
- B. AIS <6 hrs, C/I to thrombolysis
- C. AIS <6 hrs post-IV alteplase



STROKE MANAGEMENT IN LARGE VESSEL OCCLUSION (LVO)

- LVO + eligible for IVT + at MT center
 - Give IV thrombolysis (IVT) → Proceed to mechanical thrombectomy (MT)
- LVO + not near MT center
 - Give IVT (bridge therapy) → Transfer for MT
- MT not available
 - IVT alone is still beneficial
- IVT contraindicated
 - Proceed directly to MT if feasible

CONTRAINDICATIONS TO THROMBOLYSIS

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Mnemonic: STORM

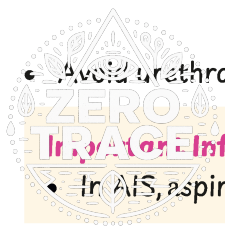
- Sustained BP >185/110 mmHg, despite treatment
- Bleeding Diathesis: INR >1.7
- Gastrointestinal Bleed : previous < 21 days
- Recent head injury / Recent ICH
- Recent MI / Major surgery in last 14 days

INDICATIONS FOR THROMBOLYSIS IN AIS

- Clinical confirmation for diagnosis of stroke (NIHSS ≥5)
- Age >18 yrs
- Window period <4.5 hrs
- CT scan: No edema/hemorrhage >1/3 MCA territory

ADMINISTRATION OF IV PA (PLASMOGEN ACTIVATOR)

- Establish 2 peripheral IV lines
- Review eligibility
- Monitor BP continuously
- Avoid antithrombotics for the next 48 hours
- Check for neuro status decline: In case of IV PA induced bleed → Give Cryoprecipitate + repeat imaging



- Avoid urethral catheterization for next 2 hrs

Information

- In AIS, aspirin is started >24-48 hours

MEDICAL SUPPORT

- Pneumatic compression stockings (to prevent DVT), UFH/LMWH after 24 -48 hrs
- BP Control threshold value
 - Thrombolysis candidate: Intervene if BP **>185/110 mmHg**
 - Non-thrombolysis candidate: Intervene if BP **>220/130 mmHg**
- Control fever, maintain blood sugar (60-180 mg/dL)
- Malignant cerebral edema:
 - Water restriction
 - Mannitol (for cytotoxic cerebral edema)
 - Craniotomy
 - Peaks by 2-3 days
- Cerebellar infarct: **Suboccipital decompression**

ETIOLOGY OF ISCHEMIC STROKE

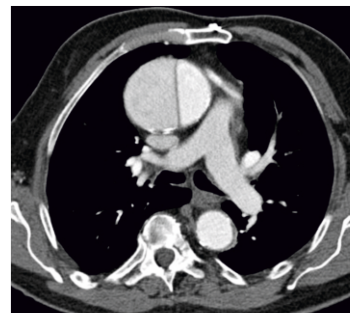
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- Cardio-embolism (20%)
 - **Non valvular AF**: Seen in structural heart disease (e.g., cardiomyopathy)
 - **Valvular AF**: D/t RHD with MS, Subacute bacterial endocarditis, Libman-Sacks endocarditis, Prosthetic heart valves
- Paradoxical embolism:
 - In DVT, a **Patent Foramen Ovale (PFO)** allows clots to bypass the lungs
 - These clots enter the left circulation and can travel to the brain, causing a paradoxical embolic stroke
- Atherosclerosis of distal ICA and proximal MCA
- ICA dissection d/t aortic dissection/ /vertebral artery dissection

Extra mile

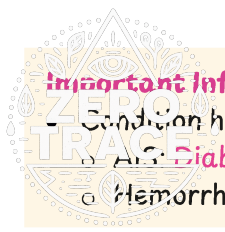
Aortic Dissection

- Only condition where chest pain at rest + neurological stroke symptoms coexist is **aortic dissection with antegrade spread involving ICA**
- **DOC** for rate control: **Esmolol IV** (↓ propagation of tear)
- Medical Management: Start with Esmolol, followed by Sodium Nitroprusside
 - If Nitroprusside is given first, it may cause reflex tachycardia
→ **↑propagation of tear**



HYPERCOAGULABLE STATES:

- Antiphospholipid Syndrome
- COVID-19
- Homocysteinemia
- Libman-Sacks Endocarditis
- Protein C, S or AT3 Deficiency
- Factor V Leiden Mutation



Important Information

- Condition having the highest relative risk for AIS: **Diabetes Mellitus**
- Hemorrhagic stroke: HTN

BUZZWORDS FOR STROKE LOCALIZATION

01:22:05

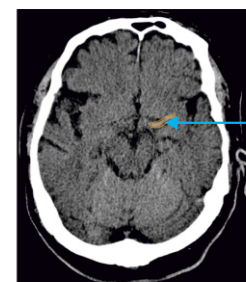
CORTICAL STROKE	• Aphasia or loss of higher mental function • Eyes deviate toward the side of the lesion
SUBCORTICAL STROKE (Internal capsule)	• Pure motor stroke
BRAINSTEM STROKE	• Cranial nerve palsy • Crossed hemiplegia • Eg: Webber syndrome (ipsilateral CN III palsy + contralateral Hemiplegia)

COMPARISON BETWEEN CORTICAL, SUBCORTICAL, AND BRAINSTEM STROKE

FEATURE	CORTICAL (MCA/ACA)	SUBCORTICAL (INTERNAL CAPSULE/THALAMUS)	BRAINSTEM (MIDBRAIN/PONS/MEDULLA)
SITE	Cerebral cortex	Internal capsule, thalamus	Midbrain, pons, medulla
CAUSE	Large artery embolism	Small vessel (lacunar) disease	Vertebrobasilar occlusion
CONSCIOUSNESS	Preserved	Preserved	May be impaired
COGNITIVE SIGNS	Aphasia, neglect	Rare	Rare
MOTOR	Contralateral weakness	Pure motor hemiparesis	Quadripareisis or crossed signs
SENSORY	Cortical sensory loss	Pure sensory stroke	Variable (may involve cranial nerves)
CN INVOLVEMENT	Absent	Absent	Present (e.g., dysphagia, diplopia)
SPEECH	Aphasia (dominant side)	Rare	Dysarthria, hoarseness

IMAGE BASED QUESTIONS

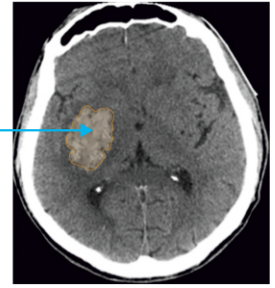
- Dense MCA Sign in AIS
 - Hypodensity in acute ischemic stroke takes 6-12 hours to develop.
 - **DWI-MRI** is preferred for early diagnosis in AIS



01:24:52
CT Showing Dense MCA sign

- Right Intracerebral Hemorrhage
 - Leading site of hemorrhagic stroke: **Putamen**
 - Leading cause: **Hypertensive crisis**
 - Target BP: **< 140/90 mm Hg**
 - Involved vessels: Penetrating branches of the lenticulostriate artery

CT Showing Hemorrhage in right hemisphere

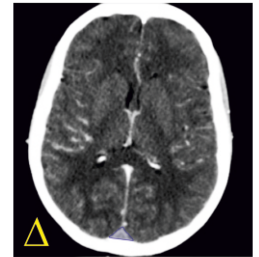


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CEREBRAL VENOUS SINUS THROMBOSIS

CAUSES:

- **Inherited Thrombophilia:** Prothrombin G20210A mutation
- Factor V Leiden mutation: defective binding of Factor-V with Protein C
- Antiphospholipid syndrome, Sickle cell anemia, Polycythemia
 - Polycythemia (↑ RBC mass) → ↑ viscosity → sluggish circulation in venous sinuses → **venous stroke**
- Protein C/S/ Antithrombin III deficiency
- Post meningitis
- **Severe dehydration** → ↑ viscosity



Empty Delta Sign

CASE SCENARIO

01:30:58

30-year-old female on OCP develops a sudden headache, new-onset seizures, focal weakness, and signs of ↑ ICP. Fundus examination shows papilledema

- ↑ Pressure in the proximal area of obstruction → vasogenic cerebral edema and possible rupture of veins causing hemorrhage
- Evaluation: **CTV/MRV → Empty delta sign**
- Rx: IV Heparin initially, then OAC (e.g., Warfarin) for 3–6 months

MOYAMOYA DISEASE

01:33:13

- Non-inflammatory occlusion of distal ICA, MCA
- Formation of collaterals which are thin-walled, can burst and cause brain hemorrhage
- In Pediatric, occlusion will cause → **Ischemic stroke**
- In Adulthood, the rupture will cause → **Hemorrhagic stroke**
- Angiography: **Puff of smoke appearance**



Puff of smoke appearance

C.A.D.A.S.I.L

01:35:16

- Cerebral Autosomal Dominant Arteriopathy with Subcortical Infarcts and Leukoencephalopathy
- Gene defect: **NOTCH3**

CASE SCENARIO

38-year-old man presents with headache episodes and is diagnosed by the physician as having **migraine with aura**. He is currently undergoing physiotherapy for an **ischemic stroke** last year that left him with left-hand weakness. At **age 30**, he had a stroke, which left him with left leg weakness. His **father died of a stroke** at the age of 60

POSTERIOR REVERSIBLE ENCEPHALOPATHY SYNDROME

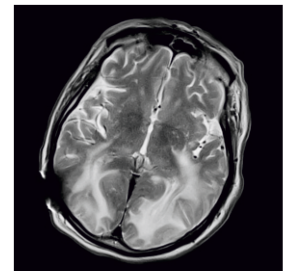
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- Vasogenic edema d/t BBB disruption from severe elevation of BP
 - Edema mainly present in occipital & parietal lobes
- Triggers:
 - Eclampsia
 - Acute kidney injury (RAAS+)
 - Amphetamine overdose
 - SLE, APS

CASE SCENARIO

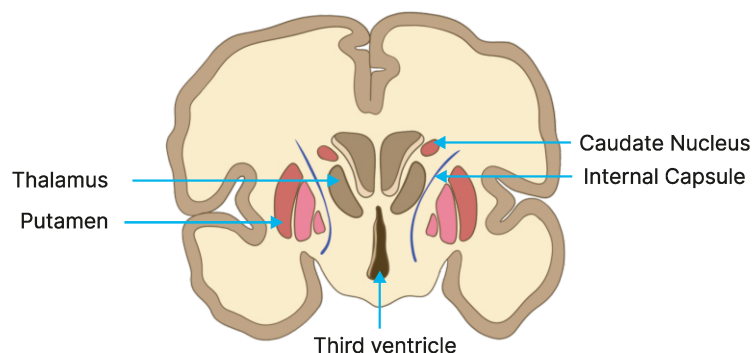
A Young female after delivery develops sudden onset headache, blindness, seizures, fluctuating neurological symptoms. FLAIR MRI shows **white matter hyperintensities** in B/L occipital and parietal lobes

- Rx: Control MAP (~130 mmHg) and control seizures with anticonvulsants



SMALL VESSEL STROKE

01:39:38



- Lacunar Stroke: Caused by Atherothrombotic occlusion / **Lipo-hyalinosis involving penetrating branches**

MANIFESTATIONS OF LACUNAR STROKE

PURE MOTOR HEMIPARESIS:

- Infarct in **posterior limb of internal capsule**

PURE SENSORY STROKE:

- **Infarct in ventral thalamus**
- **Dejerine-Roussy syndrome**: Persistent burning pain on one side not relieved by NSAIDs

ATAXIC HEMIPARESIS:

- Damage to **corticopontocerebellar tract**
- I/L ataxia + C/L hemiparesis



DYSARTHRIA-CLUMSY HAND SYNDROME:

- Affected region: **Genu of internal capsule**
- Facial weakness, dysarthria

MCQ

Q. A 65-year-old male presents to the emergency department with sudden onset weakness in his right arm and slurred speech that lasted for about 20 minutes. His blood pressure is measured at 160/100 mmHg. A risk of stroke is being calculated using ABCD² score. All of the following are true regarding this score except?

- a. Past history of stroke feature
- b. BP more than 140/90 mmHg
- c. Arm weakness
- d. Diabetes mellitus

Ans: **(a)**



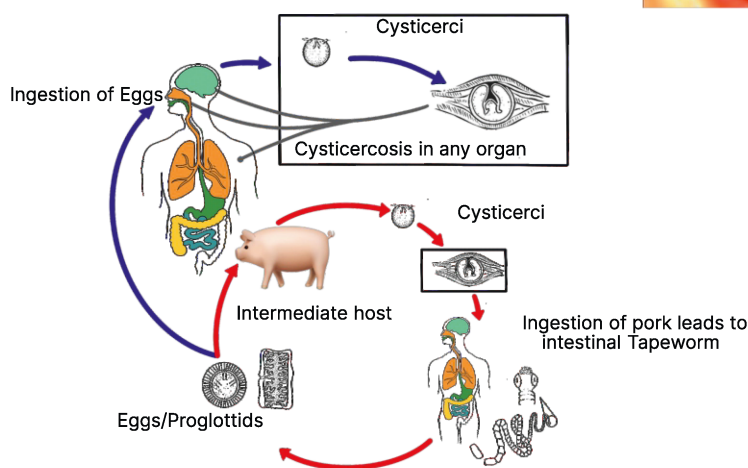
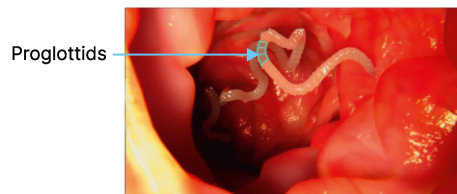
NEUROCYSTICERCOSIS AND BRAIN TUMORS



00:00:42

NEUROCYSTICERCOSIS

- Tapeworm (*Tenia solium* 3m) acquired by **semi cooked pork**
- Resides in **the upper jejunum**
- The segment in the worm → proglottids
- Each proglottid can harbor up to 10,000 eggs



- Transmission of Neurocysticercosis and Cysticercosis → **consumption of eggs**
- Consumption of proglottids/ eggs → **feco oral contamination or autoinfection**
- Smaller eggs → can cross the intestine → blood stream → muscles
- crossing BBB → **Neurocysticercosis**
- Intermediate host: pigs
- Larvae cannot cross through the intestine
- Consumption of semi-cooked pork with larva → **Intestinal tapeworm**
- Leading cause of acquired epilepsy/ seizures in India
 - Parasite in the brain → immunologic reaction → cerebral edema → seizures
 - Immediate T/t **Albendazole** can kill the parasite in the brain → leading to **calcification** → Epileptogenic focus → seizures
 - T/t with tapering of **antiepileptic drugs up to 2 years/** until the neuro symptoms subsides
- Leading cause of intracranial calcification in India
- In vegetarians → Cabbage/ Lettuce in food → If these leafy vegetables are grown in fecally contaminated soil → ingestion of eggs
- In non-vegetarians → who eat semi cooked pork, ingesting the larvae, they will form into adult tapeworms in the GI tract
- Autoinfection:
 - Poor personal hygiene: dirty nails
 - Do not wash their hands after defecating
 - More likely to ingest eggs (under nails)

TYPES OF CYSTICERCOSIS:

- **Cerebral Neurocysticercosis:** brain parenchyma → cerebritis
- **Subarachnoid cysticercosis:** Subarachnoid space, meninges, ventricular system → hydrocephalus, arachnoiditis
- **Extraneural cysticercosis:** muscles → lumps and bumps all over the body, heart, subcutaneous tissue, retina

CASE PRESENTATION

00:09:02

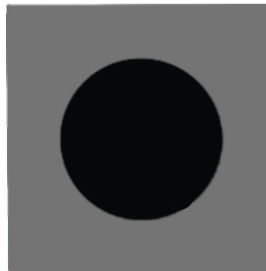
A 30 year old rickshaw puller living in slums presents with complaints of multiple episodes of GTCS/ focal seizures. On examination: GCS in 15/15; CN nerve is normal; Fundoscopy – papilledema. PE – multiple peas sized, firm nodular swellings in both upper and lower extremities and back.

Work up:

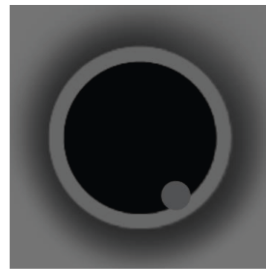
- LP- Enzyme immune transfer blot (EITB) for antibodies against cysticercosis and antigens
- Imaging modality of choice: Gadolinium-enhanced
 - **Scolex inside cyst** (hypointense lesion) → Absolute criteria
 - **Ring-enhancing lesions** (Vasogenic cerebral edema)
- Magnetic resonance spectroscopy:
 - NCC: **Amino acid peak**
 - Tuberculoma: **Lipid peak**
- Fundus examination:
 - Subretinal cysticerci
 - R/o papilledema
- X-ray of extremities:
 - **Rice grain calcification** → due to dead worms in the muscle
- NCCT head: **Starry sky appearance**

MRI BRAIN FINDINGS IN NCC

VESICULAR STAGE:



COLLOIDAL VESICULAR STAGE:



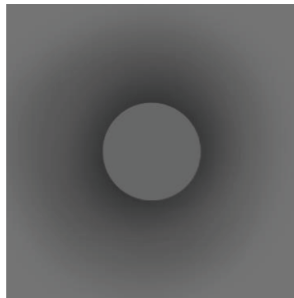
Developing scolex

Ring enhancing lesion

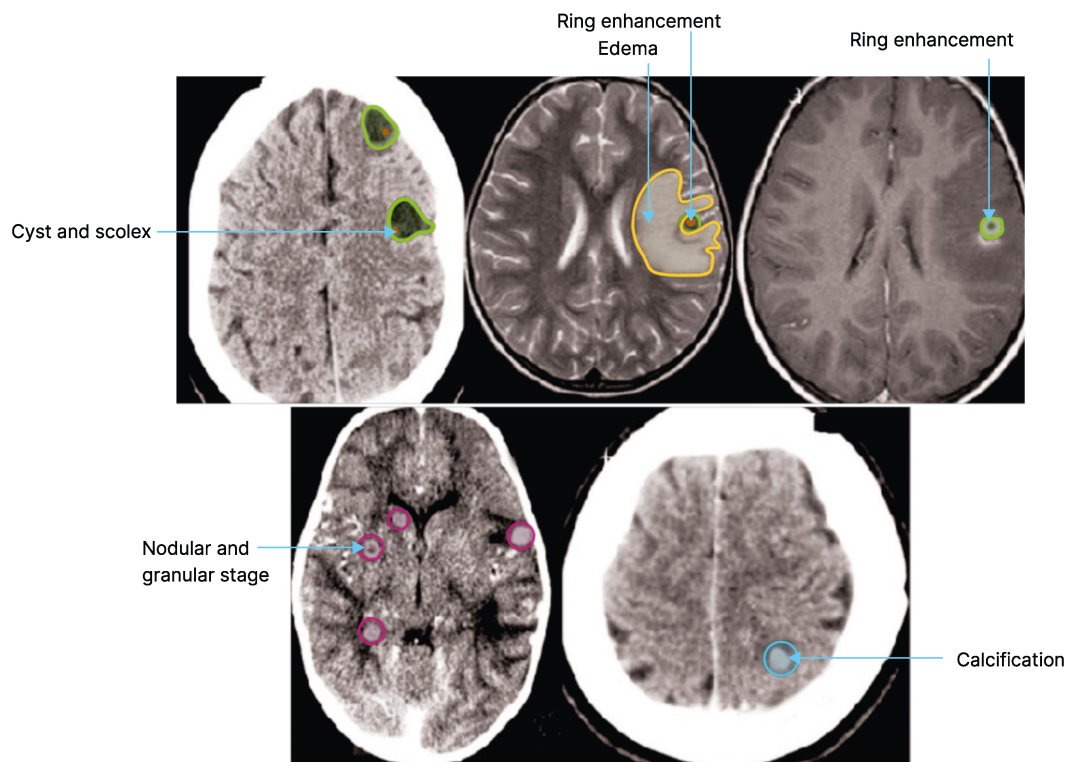
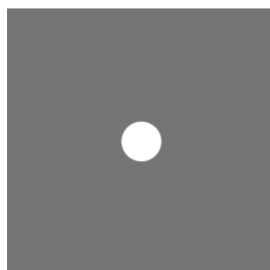


- Symptomatic phase
- Presence of substantial cerebral edema and seizures

GRANULAR NODULAR STAGE:



NODULAR CALCIFIED STAGE:



- Due to ongoing infection multiple stages can be seen in single individual

ABSOLUTE CRITERIA FOR DIAGNOSIS OF NCC (Mnemonic: HIS)

00:21:05

- Histopathological demonstration of parasite in brain/ spinal cord
- Neuroimaging showing scolex in the cystic lesion
- Subretinal cysticercosis

MAJOR CLINICAL/ EXPOSURE CRITERIA (MNEMONIC: I'M SURE OUTSIDE CONTACT)

- Detection of antibodies with EITB
- Cysticercosis **outside** CNS
- Evidence of household **contact** with *Tenia solium*

MINOR CLINICAL/ EXPOSURE CRITERIA

00:23:23

- Clinical features of NCC
- Individual living in endemic area

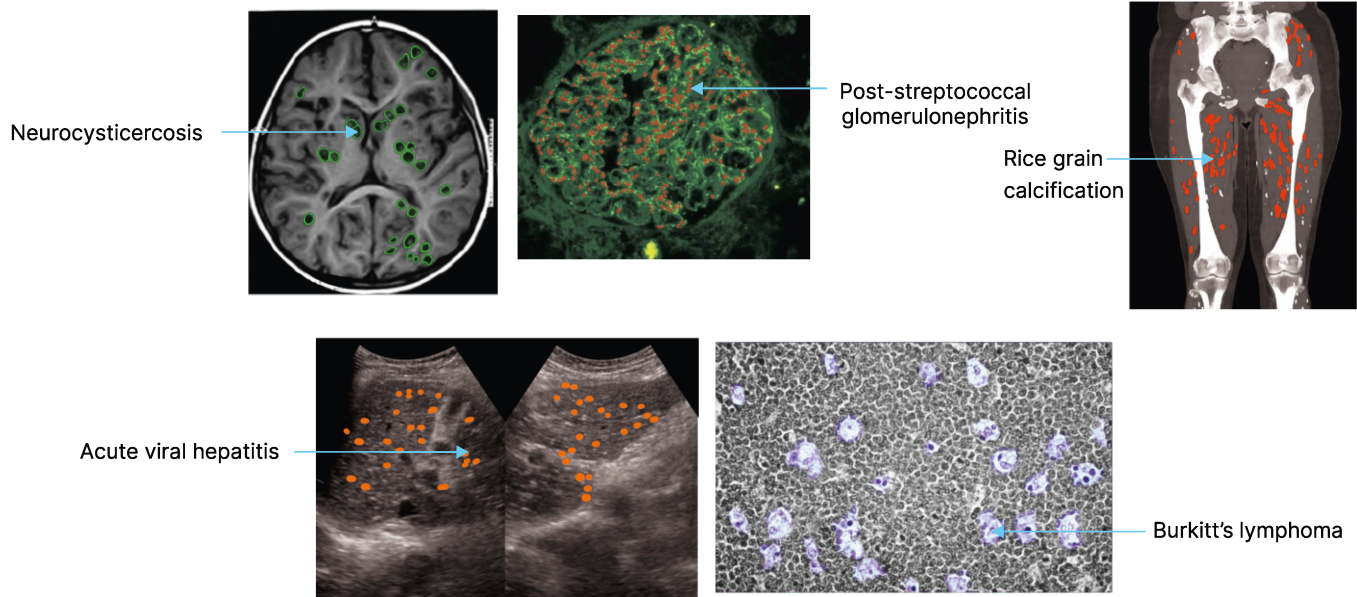
CONFIRMATIVE NEUROIMAGING CRITERIA

- Resolution of cystic lesions spontaneously or after cysticidal drug therapy

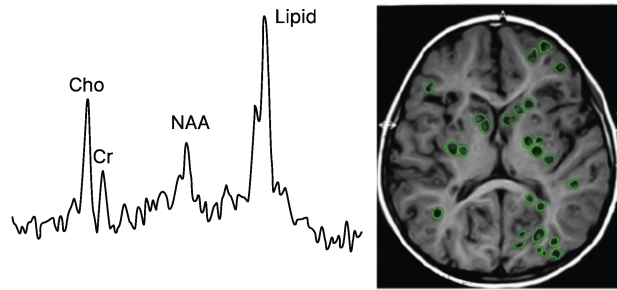
Important Information

- Diagnosis algorithm of NCC: 1 absolute criterion + 2 major criteria

STARRY SKY APPEARANCE IS SEEN IN:



- CT/ MRI head: **NCC**
- Kidney biopsy: **Post streptococcal glomerulonephritis**
- USG Liver: **Acute viral hepatitis**
- Lymph node biopsy: **Burkitt's lymphoma**
- Echocardiography/ Cardiac MRI: **Restrictive cardiomyopathy followed by amyloidosis**



- MR Spectroscopy helps or rejects a diagnosis of NCC

CYSTICERCUS GRANULOMA	TUBERCULOMA
• Round in shape	• Irregular in shape
• Cystic	• Solid
• $\leq 20\text{mm}$ / < with ring enhancement or visible scolex	• $> 20\text{ mm}$
• Cerebral edema is not enough to produce midline shift or focal neurological deficit	• Associated with severe perifocal edema and focal neurological deficit

TREATMENT OF NCC

00:29:06

- Vasogenic cerebral edema
 - I/V Dexamethasone
- Anti-parasitic drugs
 - After 48-72 hours of Dexamethasone (to $\downarrow$ the cytokine released allergic reaction/ anaphylaxis)
 - Albendazole (8 days)
 - Praziquantel (> 1 documented lesion on neuroimaging)

Important Information

- H. cyst: Albendazole for 28 days
- Anti-epileptic drugs
 - GTCS: Valproate
 - Focal seizure: Carbamazepine/ Lamotrigine
- $\uparrow\uparrow$ ICP
 - Ventriculoperitoneal shunting

} Up to 2 years (tapering dose)

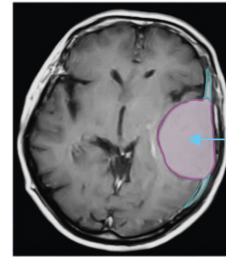
BRAIN TUMOR

00:32:24

- M/c brain tumor: Metastasis (Lungs: Oat cell cancer)
- M/c location of metastasis in Brain: Gray-white mater junction
- Causes of metastasis to the brain: Oat cell ca lung $>$ Ca breast $>$ malignant melanoma
- Tumor with highest propensity to metastasize to Brain: Malignant melanoma
- M/c 1° brain tumor: Meningioma
- Incidence wise: $2^\circ >$ Meningioma $>$ Glioma

MEMINGIOMA

- M/c primary brain tumor
- Incidence in patients exposed to cranial radiation or neuroimaging
- Derived from arachnoid cap cells/ meningothelial cells
 - Intracranial calcification
- NCCT shows
 - White lesion with **dural tail sign**
 - Well defined hyperdensity
- A/w NF-2, chromosome 22q that encodes Merlin
- Tumors seen in NF-2 are bilateral vestibular Schwannomas and Meningioma



Meningioma with Dural tail sign

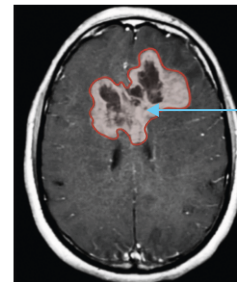
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Important Information

- NF-1: Associated with optic glioma
- Overall M/c tumor in adults due to radiation exposure:
 - Papillary CA thyroid
 - Meningioma

GRADE IV GLIOMA (GLIOBLASTOMA MULTIFORME)

- Which brain tumor in an adult have the worst prognosis?
- Ans: GRADE IV Glioma (Glioblastoma multiforme)



Butterfly Glioma

00:38:47

CLINICAL FEATURES OF BRAIN TUMOR

- Impaired cognition
- Early morning headache, holocephalic ($\uparrow$ ICP)
- Migraine like symptoms
- Personality changes : Frontal lobe
- Aphasia, arm weakness
- Seizures (+/-)

INVESTIGATION OF CHOICE:

- MRI: Gadolinium enhanced (To identify and demonstrate Vasogenic cerebral edema)
- MRS: differentiate necrosis vs tumor progression

TREATMENT

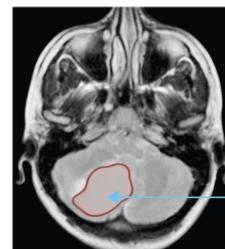
- Metastasis – Whole brain radiation (WBRT) or Stereotactic radiosurgery (SRS)
- Astrocytoma
 - Temozolomide
 - Procarbazine, Lomustine and Vincristine
- Steroids

00:43:10

VHL	• Cerebellar Ganglioblastoma
TURCOT	• Cerebellar hemangioblastoma ◦ They are not originating from cells of the cerebellum but from the blood vessels
GARDNER	• Medulloblastoma + Osteomas
NF-1	• Optic glioma
NF-2	• Meningioma
TUBEROUS SCLEROSIS	• Subependymal Astrocytoma

PEDIATRIC BRAIN TUMORS

- M/c benign brain tumor: Craniopharyngioma
- M/c malignant brain tumor: Medulloblastoma (craniospinal axis spread)
- Supratentorial tumor: Craniopharyngioma
- Tumor with worst prognosis: Brain stem glioma

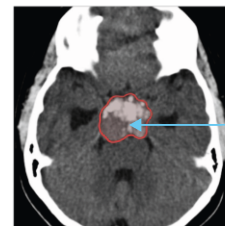


Cerebellar pilocytic astrocytoma-Hyperintense cystic lesion on T2

00:48:30

CRANIOPHARYNGIOMA

- Derived from Rathke's pouch
- Bimodal age of presentation
 - Children
 - 60 years of age
- Supratentorial/ Suprasellar tumor showing intracranial calcification



Cystic lesion with calcification in the midline

00:50:14

Clinical presentation

- Visual deficit
- Central diabetes insipidus (damage to Posterior pituitary)
 - Polyuria
 - Polydipsia
- Headache

Important Information

- Fried egg biopsy taken from stereotactic brain biopsy: **Oligodendroglioma**
- Butterfly appearance on MRI is seen in: **Grade IV glioma**
- Leading cause of Leptomeningeal spread: **CA breast**
 - Investigation: CSF cytology, MRI spine
- Leading cause of extradural spine compression: **CA breast**
 - Investigation: MRI spine